GEOMETRIC IRREDUCIBILITY OF AN IRREDUCIBLE SCHEME IS EQUIVALENT TO GEOMETRIC IRREDUCIBILITY OF ITS FUNCTION FIELD

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Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file varieties.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The proof is complete as written and ends with the local criterion applied to X; no step is referred to the reader.

Editorial note. The counted spans carry 7 navigation stubs and no bibliography citation.

Editorial note. The preamble is the chapter's retained minimal preamble (sources/stacks-varieties/retained-preamble.tex lines 1-77), a verbatim excerpt of the native preamble.

Editorial note. The label lemma-geometrically-irreducible-function-field is defined only in varieties.tex, so no cross-chapter restatement had to be weighed for this card.

Allowed standard prerequisites (editorial).

- Geometric irreducibility versus the separable algebraic closure in the field. Algebra, Lemma algebra-lemma-geometrically-irreducible-separable-elements, invoked at proof lines 1457-1459. Supplies the equivalence of conditions (2) and (3).
- Local criterion for geometric irreducibility of a scheme. Same chapter, Lemma lemma-geometrically-irreducible-local, invoked at proof lines 1464-1466 and 1481-1483. Supplies the passage between geometric irreducibility of X and geometric irreducibility of the rings of affine opens.
- Subalgebras of geometrically irreducible algebras, and the local ring versus its residue field with locally nilpotent kernel. Algebra, Lemmas algebra-lemma-subalgebra-geometrically-irreducible, algebra-lemma-minimal-prime-reduced-ring and algebra-lemma-p-ring-map, invoked at proof lines 1467-1473 and 1478-1481. Supplies the two directions of the reduction through the local ring at the generic point.

Source reference. Original lemma 0G33: sources/stacks-chow/tags.txt, source lines 20078–20078. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Source reference. Original lemma 00EU: sources/stacks-chow/tags.txt, source lines 548–548. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Date: stacks-project:varieties.tex:local-snapshot-20261011; GNU FDL 1.2 or later.

Source reference. Original lemma 0BRA: sources/stacks-chow/tags.txt, source lines 14763–14763. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Source reference. Original lemma 037N: sources/stacks-chow/tags.txt, source lines 4206–4206. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Source reference. Original lemma 038G: sources/stacks-varieties/pinned-original-source.tex, source lines 1405–1405. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Lemma 0.1. *Let X be an irreducible scheme over the field k. Let $\xi \in X$ be its generic point. The following are equivalent:*

1. *X is geometrically irreducible over k;*
2. *$\kappa(\xi)$ is geometrically irreducible over k;*
3. *the separable algebraic closure of k in $\kappa(\xi)$ is equal to k.*

Proof. By Algebra, Lemma 0G33, (2) and (3) are equivalent.

Assume (1). Recall that $\mathcal{O}_{X,\xi}$ is the filtered colimit of $\mathcal{O}_X(U)$ where U runs over the nonempty open affine subschemes of X. Combining Lemma 038G and Algebra, Lemma 037N we see that $\mathcal{O}_{X,\xi}$ is geometrically irreducible over k. Since $\mathcal{O}_{X,\xi} \to \kappa(\xi)$ is a surjection with locally nilpotent kernel (see Algebra, Lemma 00EU) it follows that $\kappa(\xi)$ is geometrically irreducible, see Algebra, Lemma 0BRA.

Assume (2). We may assume that X is reduced. Let $U \subset X$ be a nonempty affine open. Then $U = \mathrm{Spec}(A)$ where A is a domain with fraction field $\kappa(\xi)$. Thus A is a k-subalgebra of a geometrically irreducible k-algebra. Hence by Algebra, Lemma 037N we see that A is geometrically irreducible over k. By Lemma 038G we conclude that X is geometrically irreducible over k. □